

C. Interesting Sequence

time limit per test: 1 second
memory limit per test: 256 megabytes

Petya and his friend, robot Petya++, like to solve exciting math problems.

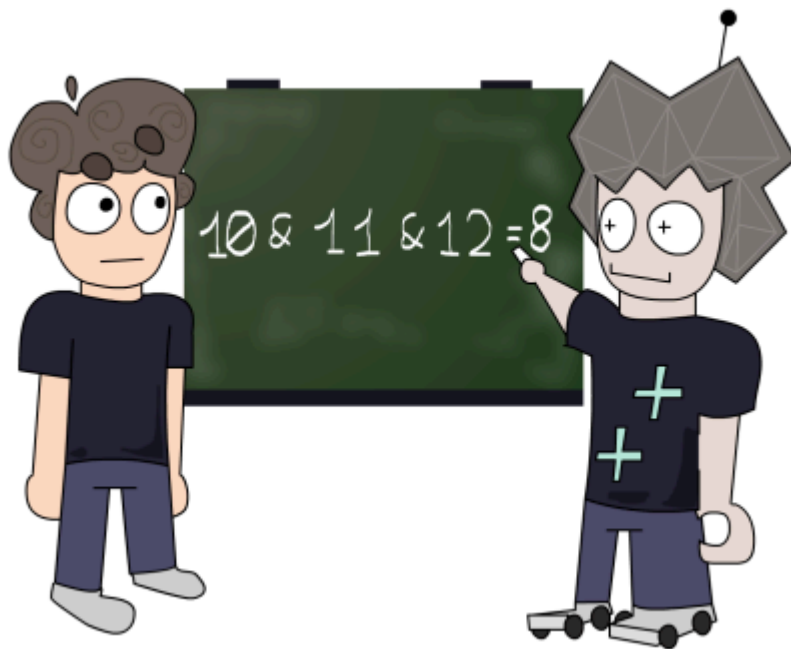
One day Petya++ came up with the numbers n and x and wrote the following equality on the board:

$$n \& (n + 1) \& \dots \& m = x,$$

where $\&$ denotes the [bitwise AND operation](#). Then he suggested his friend Petya find such a minimal m ($m \geq n$) that the equality on the board holds.

Unfortunately, Petya couldn't solve this problem in his head and decided to ask for computer help. He quickly wrote a program and found the answer.

Can you solve this difficult problem?



Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 2000$). The description of the test cases follows.

The only line of each test case contains two integers n, x ($0 \leq n, x \leq 10^{18}$).

Output

For every test case, output the smallest possible value of m such that equality holds.

If the equality does not hold for any m , print -1 instead.

We can show that if the required m exists, it does not exceed $5 \cdot 10^{18}$.

Example

input	output
5	12
10 8	10
10 10	-1
10 42	24
20 16	1152921504606846976
10000000000000000000 0	

Note

In the first example, $10 \& 11 = 10$, but $10 \& 11 \& 12 = 8$, so the answer is 12.

In the second example, $10 = 10$, so the answer is 10.

In the third example, we can see that the required m does not exist, so we have to print -1 .